TASKS 9 & 10

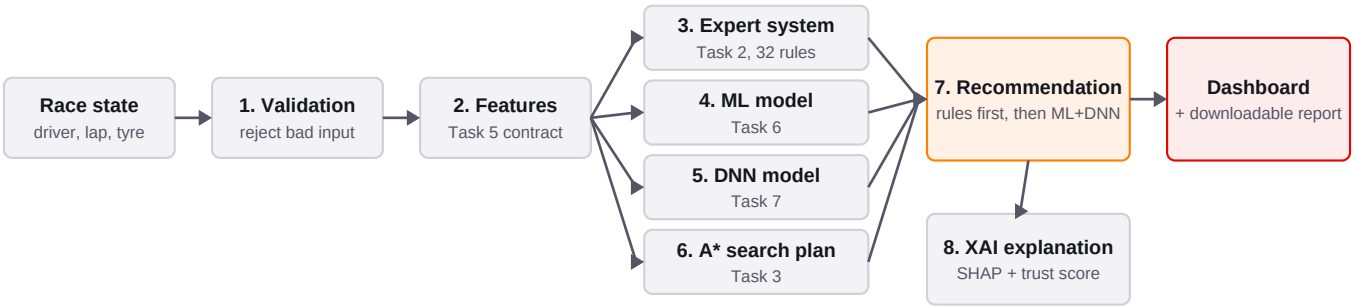
System Integration, Deployment, Evaluation & Responsible AI

F1 Race Strategy Intelligence: one real race (2023 Bahrain Grand Prix, FastF1 data, 1,055 laps, 20 drivers). Brief prepared 12 October 2026.

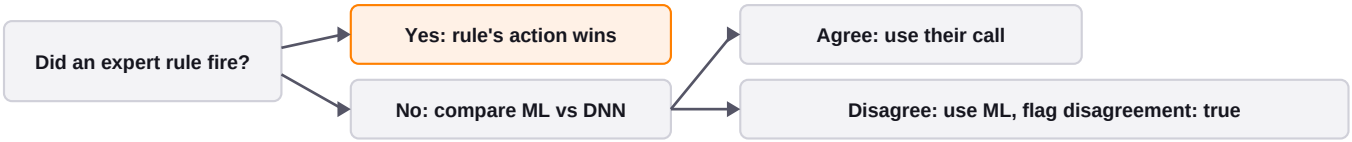
In one sentence. Task 9 joins the eight earlier tasks into one working system: a single race-state request runs the expert system, ML, deep learning, search and explainability, then returns one recommendation in a dashboard and a downloadable report. Task 10 checks that system honestly, covering correctness, speed, agreement with what real teams did, explanation quality, robustness and fairness, and reports what it cannot do.

Task 9: System Integration & Deployment

The integrated pipeline (one call: `POST /api/strategy/predict`)



How the final call is made, and why. Expert rules encode certain domain knowledge (for example, pitting under a safety car), so they take priority. Otherwise the two learned models must agree. When they disagree, the system falls back to the ML model (Task 6's selected model) and marks the result *disagreement: true* instead of hiding the conflict.



What we built and why

Component	What it is	Why we did it this way
Integrated back end	FastAPI service; `strategy_service.py` chains all stages above	Tasks 1–8 were separate scripts. One call shows they work together on the same race state.
Dashboard	Next.js web app, 8 pages: Overview, Strategy, Reasoning, Models, Explainability, Data, Model Comparison, Responsible AI	Chosen over Streamlit so the models sit behind a real HTTP API that tests and scripts also call. Every figure is read live from generated artifacts, never typed in by hand.
Report generator	One race state in; Markdown or HTML briefing out, with provenance	An engineer needs a readable record of why a call was made, not just a number.
Docker deployment	Back-end and front-end images plus `docker-compose`, with a health check	Reproducible on any machine. Models and data are mounted, not baked into images or served publicly.
Safety by design	Input validation, restricted CORS, model files not downloadable, “Not available” when an artifact is missing	A missing model must never be replaced by a fake number; bad input is rejected with a clear reason.

Task 10: Evaluation, Responsible AI & Documentation

A single script, `scripts/evaluate_system.py`, produces every result below as JSON in `artifacts/evaluation/`. Nothing is typed in by hand, so the numbers can be regenerated and checked.

1. Model results on the same held-out laps

All three models are scored on identical test data: the last 180 laps of the race (laps 47–57), never seen in training.

Task	Metric	Simple (linear)	ML (Task 6 choice)	DNN (Task 7)
Lap time (seconds)	MAE, lower is better	0.790	0.781 (SVR)	**0.565**
	R^2, higher is better	0.141	0.302	**0.456**
Pit next lap (yes/no)	PR-AUC, higher is better	**0.250**	**0.250** (random forest)	0.024
	ROC-AUC	**0.983**	**0.983**	0.777

Reading: the DNN is clearly best for lap time. For pit decisions it is worst: with only 675 training laps from one race, simpler models generalise better. The pit test window contains just **one** real pit stop, so pit test scores are fragile.

2. System evaluation

Area	How we tested	Result	What it tells us
Functional testing	Automated test suite (pytest)	**307 passed, 0 failed** (12 Oct); 64.5% code coverage at the 1 Oct evaluation run	End-to-end prediction, input rejection, missing-model handling and the report all work.
Performance	200 timed API requests	Median **101 ms**, 95th percentile 125 ms, 0 errors; with live SHAP explanation, median 258 ms	Fast enough for interactive use, lap by lap.
Domain validation	Replayed every driver's real laps (934) through the live engine and compared pit calls with what teams actually did	Caught **34 of 39** real stops (recall 0.87); precision 0.125 (237 extra calls)	It rarely misses a stop but over-calls. This measures agreement with the teams, not optimality, and the replay includes laps the models trained on.
Expert-system check	Compared safety-car rule firings with the real track-status history	8 firings, **all** on real safety-car / VSC laps	No false safety-car calls.
Rule reachability	Checked which rule inputs the live pipeline supplies	**10 of 32** rules can never fire	They need rival-car data (gaps, undercut threat) that a single-car snapshot lacks.
Explainability	200 bootstrap resamples; feature-ablation test	Top SHAP feature stable in 92% (lap time) and 100% (pit) of resamples; removing it moves predictions more than removing a random feature	Explanations are stable and actually reflect what drives the model.
Deployment readiness	Docker files, health check, tests, docs	**5 of 5** checks pass; 12 models served	The system can be stood up and monitored.

3. Responsible AI

Also shown live in the dashboard's **Responsible AI (RAI)** tab, which compares all three models.

Principle	What we did	Finding
Robustness	Nudged the race state (track temperature ±5 °C, tyre age ±2 laps) and counted how often the call flipped	Cooler track or 2-lap-fresher tyres flipped a STAY OUT call in **7 of 9** trials each; warmer or older: 0 flips. Calls near the boundary are fragile, so a human must review them.
Fairness	Looked for protected attributes; compared accuracy across teams, tyre compounds and drivers for all three models	There are **no** demographic attributes in F1 timing data, so demographic fairness is reported as **Insufficient Data** rather than invented. By compound, every model is worse on SOFT than HARD tyres (DNN: 0.78 s vs 0.39 s). McLaren's 4x error comes from one recorded out-lap, flagged as a small sample.
Confidence	Probability outputs, plus Monte Carlo dropout (100 passes) for the DNN	DNN's 1-sigma / 2-sigma bands contain the real lap 79% / 94% of the time (68% / 95% expected), but its uncertainty does not track its error (rank correlation -0.01). The calibration curve is withheld: one pit stop is too few.
Transparency	Model cards for each model, built from the registry and metrics	Purpose, data, preprocessing, results, strengths, limitations, risks and version, all from real records.
Human oversight	Stated in every report and model card	“Decision support for engineers, not an autonomous strategy system, and not for betting.”

4. Documentation delivered

SRS, SDD, user manual, testing report, deployment guide, final Task 9/10 report (.docx) and a deliverables checklist showing **55 of 55** required items built, each opened and checked rather than ticked by filename.

Honest limitations (and why we report them)

- **One race only.** Results describe the 2023 Bahrain GP and should not be generalised to other circuits or seasons.
- **One real pit stop in the test window.** Pit-decision test scores can change completely on a single lap; cross-validation (36 pit laps) is more informative.
- **The lap-time DNN overfits.** Its training curves show a large train/validation gap; early stopping restored epoch 10 of 35.
- **Pit calls are over-eager** (precision 0.125) and **10 of 32 rules are unreachable** without rival-car data.
- **Why say all this?** Our design rule throughout: a value that cannot be produced from the real system is shown as “Not available” or “Insufficient Data”, never estimated to look complete.

Likely questions, short answers

Why does the DNN win lap time but lose pit decisions?	Lap time is a smooth, data-rich signal. Pit stops are rare events (36 of 635 cross-validation laps, about 1 in 18), and 675 laps are too few for a network to learn them reliably.
Why not demographic fairness?	The data has no protected attributes. Inventing them would be fabrication, so we measure performance gaps across operational groups (team, tyre, driver) instead.
How do you know the dashboard numbers are real?	Every page reads generated artifacts at request time, and an automated test recomputes predictions from the saved models and checks they match the stored metrics.
What would you do next?	Train across many races, add rival-car data so the remaining rules can fire, and use a live timing feed instead of a typed-in race state.